INTERNSHIP TRACKER

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

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ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**INTERNSHIP TRACKER**" is the bonafide work of ARUNTHIRAVIAM M (2116241501023), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026-2027, Semester V, under my supervision.

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DECLARATION

I, ARUNTHIRAVIAM M (2116241501023), hereby declare that the project report titled "**INTERNSHIP TRACKER**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report – including its data model, forms, reports, dashboard and permission profiles – was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

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ARUNTHIRAVIAM M

ABSTRACT

Internships are a compulsory part of an engineering programme, yet the way a college follows them is often informal. Students apply to companies by e-mail or messages, progress is reported in chats, the internship status is updated by hand, and the coordinator has no single place to see who has applied where, who is working on which tasks and who has finished. This makes it hard to monitor students, to verify dates and to prepare the internship records that are needed at the end of the semester.

To address this problem, the Internship Tracker was developed on Zoho Creator, a low-code application platform. The application has two forms. The Internship_Application form records the student name, e-mail, phone number, company, role, domain, application date, internship type, work mode, stipend, application status (default "Applied"), job description and job summary. The Internship_Progress form is linked to the application through a lookup field and records the start and end dates, the internship status, the tasks completed and the total tasks, the progress percentage, the mentor name, the weekly report and the final feedback. Two list reports present the records of the two forms, and a dashboard page shows the total tasks, the average progress, the total stipend, the work-mode count and charts. Five form workflows written with Deluge enforce the rules of the process: one validates the application on submission, one verifies the dates of a progress record, one recalculates the progress when the tasks completed are entered, and two run after a successful submission to calculate the progress and to set the internship status to Completed or In Progress.

The application was tested with 17 functional test cases on live records of three applications and five progress records. The results show that applications and progress are captured through mandatory, validated fields and that the status follows the task counts automatically. The observations also show that the company and role of the application form are still free text, which allows inconsistent spellings; replacing them with drop-downs and adding e-mail notifications, filters and an approval process are proposed as the next enhancements.

Keywords: Low-Code Development; Zoho Creator; Internship Management; Deluge Workflows; Progress Tracking; Dashboard

PROJECT AT A GLANCE

Project Title	Internship Tracker
Application Domain	Education / Internship and Placement Management
Student Name & Register No.	ARUNTHIRAVIAM M – 2116241501023
Zoho Creator Edition	Trial Edition
Application Name (Link Name)	Internship Management System (internship-management-system); project title: Internship Tracker
Published / Live URL	https://creatorapp.zoho.in/arunthiraviam.m.2024.aiml420/internship-management-system/#Section_1
Demo Credentials (per role)	Admin
Number of Forms / Subforms	2 (Internship_Application, Internship_Progress) / 0
Number of Reports (by type)	List: 2 | Kanban: 0 | Calendar: 0 | Pivot/Chart: 0 | Others: 0
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form: 5 | Schedule: 0 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	5 workflow actions; Deluge script of Update_Internship_Progress documented in Section 5.5 and Appendix C
Integrations	None in version 1.0 (e-mail notification planned)
Roles & Profiles Defined	Default Zoho Creator permission profiles
Environments Used	Development
Date of Production Deployment	20/10/2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

An internship gives an engineering student the chance to apply classroom knowledge in a company. For the college, an internship involves a series of steps: the student applies to a company for a role, the internship starts on an agreed date, the student works on a set of tasks under a mentor, a weekly report is prepared, and the internship ends as completed, on hold or discontinued. The coordinator must be able to follow every student through these steps.

In many colleges this follow-up is done through messages, phone calls and spreadsheets. Each student reports in a different way, the information is scattered, and nobody can say at a glance which students have applied, which are working and how far each internship has progressed. The effort to collect and compile the information grows with the number of students.

This project builds the Internship Tracker, a single application on Zoho Creator in which a student records the internship application and the progress of the internship, in which the status of the internship is calculated automatically from the tasks completed, and in which a dashboard gives the coordinator a summary of the work at a glance.

1.2 Problem Statement

Colleges lack a simple, central system that records internship applications and tracks the progress of each internship. Because applications and progress reports are scattered across messages and files, it is difficult to verify dates, to know the real status of an internship and to prepare reports. An application is needed that captures each fact once, validates it and keeps the status consistent with the work done.

- Application details (company, role, domain, type, mode and stipend) are collected in different formats.
- Start and end dates are not verified, so wrong dates enter the records.
- Progress is reported in free-form messages and is not expressed as a number or a percentage.
- The internship status is updated by hand and is often outdated.
- The coordinator must compile the lists and reports manually.

1.3 Objectives

- To design two linked forms that capture internship applications and internship progress with mandatory, validated fields.
- To automate the progress percentage and the internship status using Deluge form workflows.
- To verify the entered data through validation workflows on form submission.
- To provide list reports and a dashboard that give the coordinator an up-to-date view of applications and progress.
- To test the application, record the defects found and define the improvements for the next version.

1.4 Scope of the Project

In scope. The application covers two modules – Internship Application and Internship Progress – each with a data-entry form and a list report, a dashboard page with summary figures and charts, and five form workflows that validate the data and update the progress and status. The Internship_Progress form is linked to the Internship_Application form through a lookup field. Dates use the dd-MMM-yyyy format. The application is accessible on the web and through the responsive Zoho Creator interface.

Out of scope. E-mail and calendar notifications, document upload of offer letters, integration with external job portals and approval of applications by the coordinator are not part of version 1.0. They are discussed as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

Low-code development platforms provide visual environments in which data models, forms, reports and workflows are assembled by configuration, with code needed only for specialised logic [1], [2]. Research comparing low-code with model-driven engineering notes that such platforms run on the cloud, reduce manual coding and are aimed at users who are not professional programmers [3]. These characteristics match the needs of this project.

The internship process has a small and well-understood data model, the users are students and a coordinator, and the rules (date checks, progress calculation and status update) are short. Zoho Creator supplies hosting, a managed database, form, report and page builders and the Deluge scripting language, so the whole application could be built without any infrastructure. The rules were written as a few lines of Deluge instead of a back-end program, and the college can change them without redeploying code.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing arrangement a student finds an internship and informs the college by e-mail or message. The details are noted in a spreadsheet or on paper. During the internship the mentor or coordinator asks for progress by phone or chat, and the student replies in free-form messages. The status of the internship is updated by hand, and at the end of the semester the coordinator compiles the report manually. Figure 2.1 shows this as-is process.

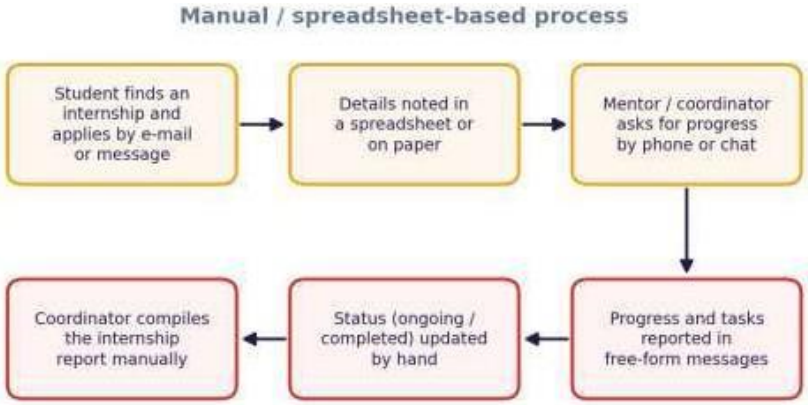


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Application details are collected in different formats and are incomplete or inconsistent.
- Dates are not verified, so an end date earlier than the start date can go unnoticed.
- Progress is not measured: there is no count of tasks completed against the total tasks and no percentage.
- The internship status depends on someone remembering to update it.
- Anyone who holds the spreadsheet can see and change the contact details of every student.
- Reports must be compiled by hand, which delays decisions.

2.3 Low-Code versus Traditional Development

Low-code platforms are reported to shorten delivery because they automate much of the work of building and deploying business applications [4]. Table 2.1 compares the two approaches for this application. The effort figures are estimates made by the author for a comparable two-form application and are given only to illustrate the difference in order of magnitude.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 4–6 weeks for database design, back-end, front-end and authentication	Estimated 1–2 days to configure 2 forms, 2 reports, 1 dashboard and 5 workflows
Skills required	Web development, database design, server administration	Data modelling and basic platform configuration; Deluge for the rules
Hosting & infrastructure	Server, database and domain must be arranged and maintained	Hosted and managed by the platform
Mobile application	Separate mobile app or responsive front-end must be built	Available through the Zoho Creator mobile interface without extra coding
Validation & business rules	Written and tested in back-end code	Form workflows with a few lines of Deluge
Maintenance & change	Code change, testing and redeployment for every new field	A field or a rule is changed in the builder in minutes
Customisation limits	Almost unlimited	Bounded by platform features; pro-code extensions needed for complex needs
Cost	Developer time plus hosting costs	Platform subscription; no separate hosting cost

2.4 Literature Survey

Table 2.2 summarises the sources reviewed for this project: textbooks and platform documentation on Zoho Creator and Deluge, research papers on low-code platforms, and an existing product in the domain of student internship and event management.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide	Explains low-code concepts, the role of the citizen developer and a step-by-step approach to building applications	Course text; guided the overall development approach
2	Zoho Corporation [2], 2026	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, permission profiles and deployment in Zoho Creator	Primary reference for configuring the application
3	D. Di Ruscio et al. [3], 2022	Low-code development and model-driven engineering: Two sides of the same coin?	Compares low-code with model-driven engineering; notes cloud execution, reduced hand coding and strengths and weaknesses	Supports the choice of a low-code platform and the discussion of its limits

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
4	R. Sanchis et al. [4], 2020	Low-code as enabler of digital transformation in manufacturing industry	Shows that low-code platforms automate application development and support digital transformation	Supports replacing manual tracking with a low-code application
5	Zoho Corporation [5], 2026	Deluge Script Reference Guide	Defines the Deluge scripting language, including conditions, operators and field assignments	Basis for the workflow scripts
6	A. Sahay et al. [6], 2020	Supporting the understanding and comparison of low-code development platforms	Presents a framework to classify and compare low-code platforms	Helps position Zoho Creator among low-code platforms
7	Zoho Corporation [7], 2026	Zoho Creator Learning Centre	Tutorials and sample applications for building Zoho Creator apps	Reference for form and workflow design practice

2.5 Emerging Trends in Low-Code Development

Several trends are shaping low-code development. AI-assisted application building allows a first version of an application to be generated from a plain-language description and then refined by hand. Citizen development lets domain experts such as a placement coordinator build and change their own tools. Hyper-automation combines workflows, scripts and integrations so that routine steps happen without manual effort, and composable applications connect small applications through APIs instead of building one large system [3], [6].

These trends influenced this project in two ways. The application was kept small and clear so that a coordinator can understand and extend it. The rules that would otherwise be done by hand – date verification, progress calculation and status update – were automated with form workflows, which is an example of hyper-automation on a small scale.

2.6 Proposed System

The proposed system is the Internship Tracker built on Zoho Creator. It replaces messages and spreadsheets with two forms, two reports and a dashboard. Each limitation listed in Section 2.2 is addressed as follows. Application details are captured in a fixed set of mandatory fields with drop-downs for the domain, work mode and status and options for the internship type. Dates are verified by a validation workflow. Progress is recorded as tasks completed against total tasks, and the progress percentage is calculated by workflows. The internship status is set automatically to Completed when all tasks are done and to In Progress otherwise, unless the internship is on hold or stopped. The reports give the coordinator a current list of applications

and of progress at any time, and the dashboard summarises the tasks, the average progress, the stipend and the work modes.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1 lists the stakeholders of the application and their responsibilities.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Student	Undergraduate student doing an internship	Submit the internship application; enter progress, tasks completed and weekly report
Mentor	Faculty member or company mentor who guides the student	Review progress and weekly reports; confirm the status of the internship
Coordinator	Faculty member in charge of internships	Monitor all applications and progress through the reports and the dashboard
Administrator	Owner of the application	Manage users and permission profiles; configure forms, reports, pages and workflows

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall record an internship application with student name, e-mail, phone, company, role, domain, application date, internship type, work mode, stipend, application status, job description and job summary.	Student, Administrator	High
FR-02	The system shall record internship progress against an internship application (selected through a lookup) with start date, end date, internship status, tasks completed, total tasks, progress percentage, mentor name, weekly report and final feedback.	Student, Mentor	High
FR-03	The system shall set the internship status automatically: Completed when tasks completed reach the total tasks, otherwise In Progress, unless the status is On Hold or stopped.	System	High
FR-04	The system shall calculate the progress percentage from the tasks completed and the total tasks.	System	High
FR-05	The system shall validate the entered data on form submission, including the internship dates.	System	High
FR-06	The system shall provide a list report for the applications and a list report for the progress records.	All roles	High
FR-07	The system shall allow a record to be searched, viewed, edited, duplicated or deleted from a report according to the permission profile.	Coordinator, Administrator	Medium
FR-08	The system shall display dates as dd-MMM-yyyy and the stipend with thousands separators and two decimal places.	All roles	Low

FR ID	Requirement	Role(s)	Priority
FR-09	The system shall provide a dashboard with the total tasks, the average progress percentage, the total stipend, the number of progress records, the work-mode count and quick links to the forms and reports.	Coordinator, Mentor	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 1,000 records shall load within 5 seconds on a normal broadband connection.
NFR-02	Security	Users shall access only the data permitted by their permission profile; contact details shall be protected.
NFR-03	Usability	The application shall be usable on desktop, tablet and mobile browsers and through the Zoho Creator mobile interface.
NFR-04	Availability	The application shall be available whenever the Zoho Creator cloud service is available.
NFR-05	Maintainability	A new field or rule shall be added through the builder without redeploying code.
NFR-06	Data quality	Required fields shall be mandatory and the dates shall be verified before a record is saved.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Trial Edition at the time of development)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	None in version 1.0
Mobile	Zoho Creator mobile app (Android / iOS) or a mobile browser
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases of the application for its two main actors, the Student and the Mentor or Coordinator. Tables 3.5 to 3.7 describe three key use cases in detail.

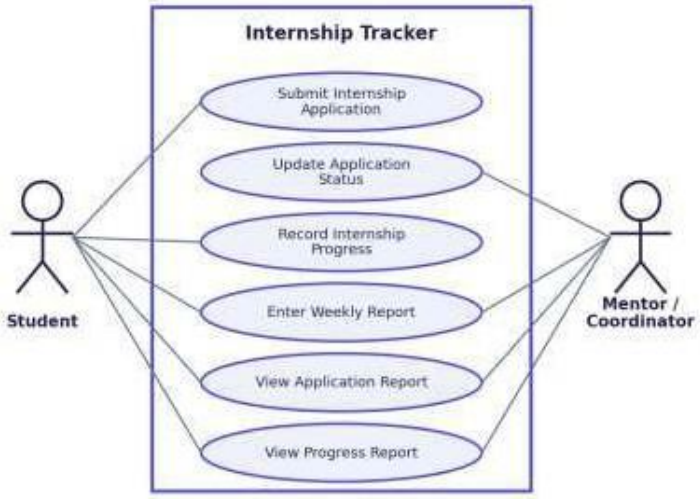


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit Internship Application

Use Case ID	UC-01
Use Case Name	Submit Internship Application
Actor(s)	Student, Administrator
Pre-condition	The user is logged in and has permission to add records to the Internship_Application form.
Main Flow	1. The user opens the Internship_Application form. 2. The user enters the student name, e-mail, phone, company, role and domain. 3. The user enters the application date, internship type, work mode, stipend and job description. 4. The application status shows Applied by default. 5. The user clicks Submit; the validation workflow runs and the record is saved.
Alternate Flow	If a mandatory field is empty or a value is invalid, the system shows an error and the record is not saved.
Post-condition	The application appears in the Internship_Application Report.
Related FR	FR-01, FR-05, FR-06

Table 3.6: Use Case Description – Record Internship Progress

Use Case ID	UC-02
Use Case Name	Record Internship Progress
Actor(s)	Student, Mentor
Pre-condition	The internship has started and the application is recorded.

Main Flow	1. The user opens the Internship_Progress form. 2. The user selects the internship application from the lookup and enters the start date and end date. 3. The user selects the internship status and enters the tasks completed and the total tasks; the progress percentage is updated. 4. The user enters the mentor name and the weekly report and clicks Submit. 5. The Date_Verify workflow checks the dates and, after the save, the success workflows set the progress and status.
Alternate Flow	If the dates are invalid, the validation workflow stops the submission and the user corrects the dates.
Post-condition	The progress record appears in the Internship_Progress Report with the updated status.
Related FR	FR-02, FR-03, FR-04, FR-05

Table 3.7: Use Case Description – Monitor Internships

Use Case ID	UC-03
Use Case Name	Monitor Internships
Actor(s)	Mentor, Coordinator
Pre-condition	Applications and progress records exist.
Main Flow	1. The user opens the Dashboard to see the summary figures and charts. 2. The user opens Internship_Application Report to see the applications. 3. The user opens Internship_Progress Report to see the dates, status and tasks completed of every student. 4. The user searches a record or opens it to read the weekly report.
Alternate Flow	If a record needs correction, the user edits it from the report and the workflows run again on submission.
Post-condition	The coordinator knows the current status of every internship.
Related FR	FR-06, FR-07, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The solution follows the layered architecture shown in Figure 4.1. Students, mentors, coordinators and administrators reach the application through a desktop browser, a tablet or mobile browser, or the Zoho Creator mobile app. The Zoho Creator application layer contains two forms, two list reports, a dashboard page and five form workflows written in Deluge. All records are stored in the managed Zoho Creator cloud database, and dates are shown in the dd-MMM-yyyy format. The integrations shown with a dashed outline are planned and are not part of version 1.0.

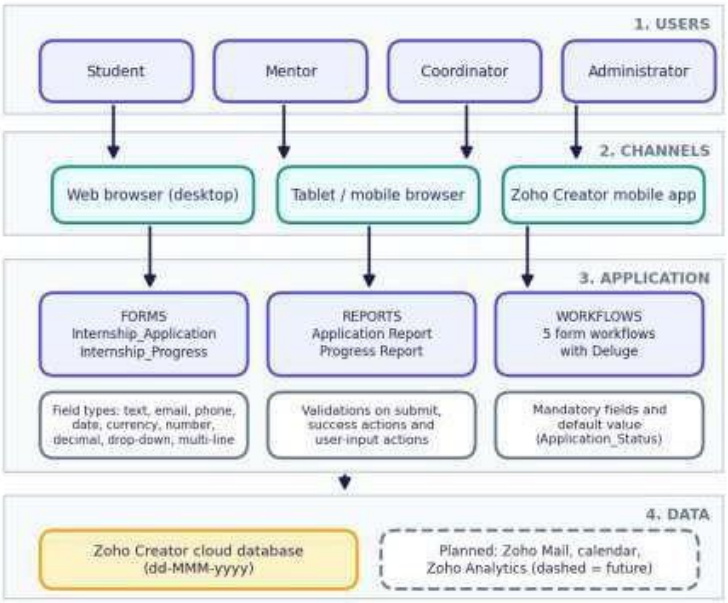


Figure 4.1: System Architecture of the Internship Tracker

Users layer. The four roles use the application according to their permission profile. **Channel layer.** The responsive Zoho Creator interface adapts to desktop, tablet and mobile screens. **Application layer.** The forms capture data, the reports and the dashboard present it and the workflows enforce the rules. **Data layer.** The records of the two forms are stored in the platform database.

4.2 Data Model

The data model has two entities, Internship_Application and Internship_Progress, shown in Figure 4.2 and Table 4.1. An application describes the internship that a student has applied for, and a progress record describes the work done during that internship. The two forms are linked

by a lookup field: the Internship_Progress form has an Internship Application field in which the user selects the application the progress belongs to, so the student, company and role are not typed again. No subform is used in version 1.0.

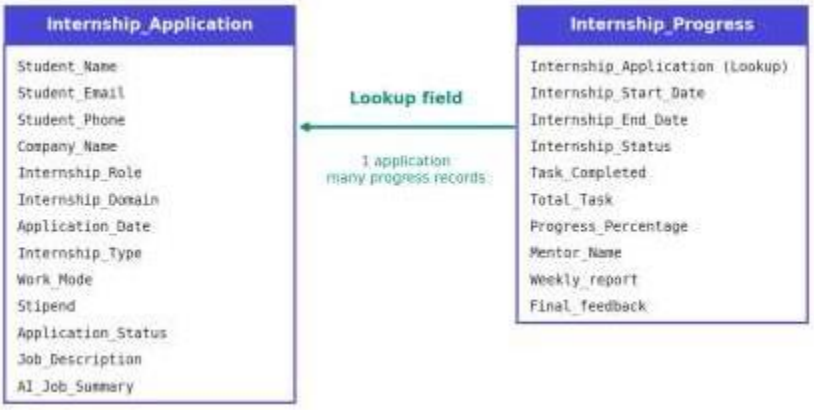


Figure 4.2: Entity–Relationship Diagram of the Internship Tracker

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type
Internship_Application	Student_Name, Company_Name, Internship_Role, Internship_Domain, Application_Status	Internship_Progress	One-to-Many (one application, many progress updates)
Internship_Progress	Internship_Application (lookup), Internship_Status, Task_Completed, Total_Task, Progress_Percentage	Internship_Application	Many-to-One (through the lookup field)

4.3 Form Design

Tables 4.2 and 4.3 give the field specification of the two forms as seen in the live application. Zoho Creator field types are named as they appear on the platform. Eleven of the thirteen fields of the application form are mandatory; Application_Status has a default value and AI_Job_Summary is optional. In the progress form the Internship_Application lookup, the two dates, the status and Total_Task are mandatory, and Task_Completed, Progress_Percentage, Mentor_Name, Weekly_report and Final_feedback are optional.

Table 4.2: Field Specification – Internship_Application Form

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Student_Name	Name	Yes	–	First name box

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Student_Email	Email	Yes	Email format	Shown as a mail link in the report
Student_Phone	Phone	Yes	Country code selector (+91)	Shown as a call link
Company_Name	Single Line	Yes	–	–
Internship_Role	Single Line	Yes	–	Free text
Internship_Domain	Drop Down	Yes	Fixed choices (for example Data Science, Web Development, Artificial Intelligence, UI/UX Design)	–
Application_Date	Date	Yes	dd-MMM-yyyy	–
Internship_Type	Radio	Yes	Paid / Unpaid / Other	–
Work_Mode	Drop Down	Yes	Fixed choices (On-site, Hybrid, Remote)	Colour-coded in the report
Stipend	Currency	Yes	Thousands separators, two decimals	Shown with the $ symbol
Application_Status	Drop Down	No	Default: Applied	–
Job_Description	Multi Line	Yes	–	–
AI_Job_Summary	Multi Line	No	–	Summary of the job description

Table 4.3: Field Specification – Internship_Progress Form

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Internship_Application	Lookup	Yes	Selects a record of Internship_Application	Replaces the typed student, company and role
Internship_Start_Date	Date	Yes	dd-MMM-yyyy	Checked by Date_Verify
Internship_End_Date	Date	Yes	dd-MMM-yyyy	Checked by Date_Verify
Internship_Status	Drop Down	Yes	Choices include Not Started, In Progress, Completed, On Hold	Set by workflow; colour-coded in the report
Task_Completed	Number	No	–	Triggers a user-input workflow
Total_Task	Number	Yes	–	–
Progress_Percentage	Decimal	No	Two decimal places	Calculated by workflow
Mentor_Name	Single Line	No	–	–

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Weekly_report	Multi Line	No	–	–
Final_feedback	Multi Line	No	–	–

4.4 Process and Workflow Design

Figure 4.3 shows the process that the application supports. The student submits the application, which is validated and saved with the status Applied. When the internship begins, a progress record is created. On every submission the Date_Verify workflow checks the dates, the progress percentage is calculated, and the status is set to Completed when the tasks completed reach the total tasks and to In Progress otherwise.

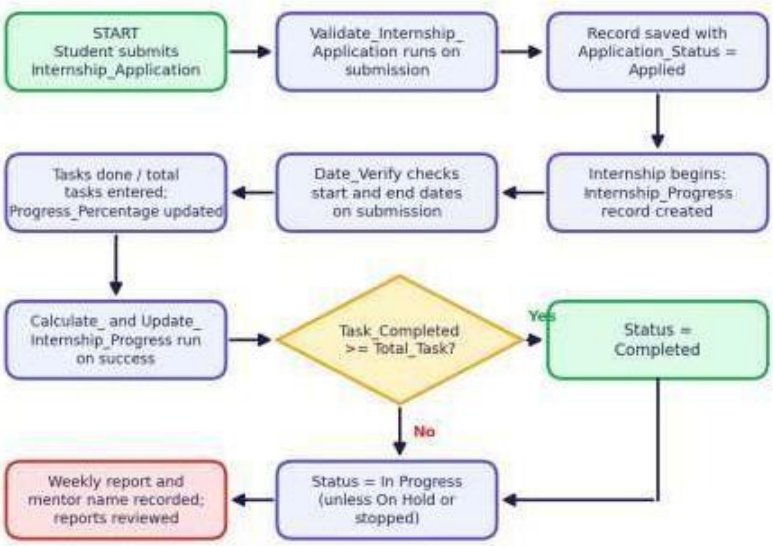


Figure 4.3: Workflow / Process Flow of the Internship Tracker

4.5 User Interface and Navigation Design

The navigation has three entries in the top menu, as shown in Figure 4.4: the Dashboard, Internship Progress and Internship Applications. The two form entries each contain the data-entry form and its report, so a user finds both in one place, and the Dashboard summarises both. The forms use left-aligned labels in a single column and the Submit and Reset buttons are at the end of the form.

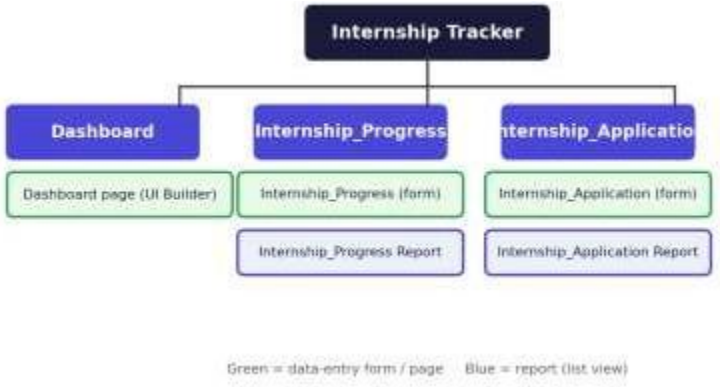


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.4 defines the permission matrix that the application is designed to enforce. A = Add, E = Edit, V = View, D = Delete, – = No access.

Table 4.4: Role–Permission Matrix

Component	Administrator	Coordinator	Mentor	Student
Form: Internship_Application	A E V D	A E V	V	A
Form: Internship_Progress	A E V D	A E V	A E V	A
Report: Internship_Application Report	V	V	V	V (own)
Report: Internship_Progress Report	V	V	V	V (own)
Page: Dashboard	V	V	V	–

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in a Zoho Creator workspace and appears in the application header as “Internship Management System”; the project is reported under the title Internship Tracker. The top menu has three entries – Dashboard, Internship Progress and Internship Applications – as seen in Figures 5.1 to 5.6. Dates use the dd-MMM-yyyy format. Table 5.1 summarises the configuration.

Table 5.1: Application Configuration Summary

Application name	Internship Management System (project title: Internship Tracker)
Method of creation	From scratch
Date format	dd-MMM-yyyy
Application menu	Dashboard (page), Internship Progress (form + report), Internship Applications (form + report)
Form label placement	Left
Preview modes	Desktop, tablet and mobile previews are available from the application toolbar

5.2 Forms and Data Capture

Two forms capture the data of the application. The field specifications are given in Tables 4.2 and 4.3. Twelve distinct field types are used: Name, Single Line, Multi Line, Email, Phone, Drop Down, Radio, Date, Currency, Number, Decimal and Lookup.

Internship_Application form. The form in Figures 5.1 and 5.2 collects the student details, the company and role, the domain, the application date, the internship type, the work mode, the stipend, the application status, the job description and the job summary. The phone field offers a country-code selector with +91 pre-selected, the internship type is chosen with radio buttons (Paid, Unpaid, Other), and the stipend field shows a numeric placeholder. The Submit and Reset buttons are at the end of the form.

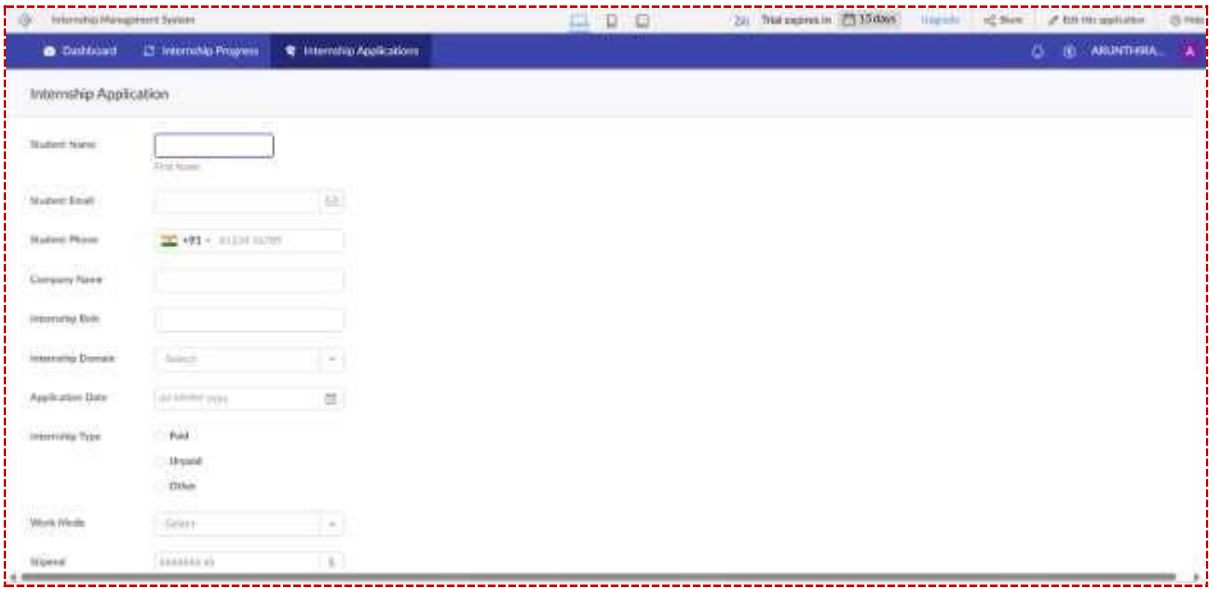


Figure 5.1: Internship_Application Form – Student and Internship Details

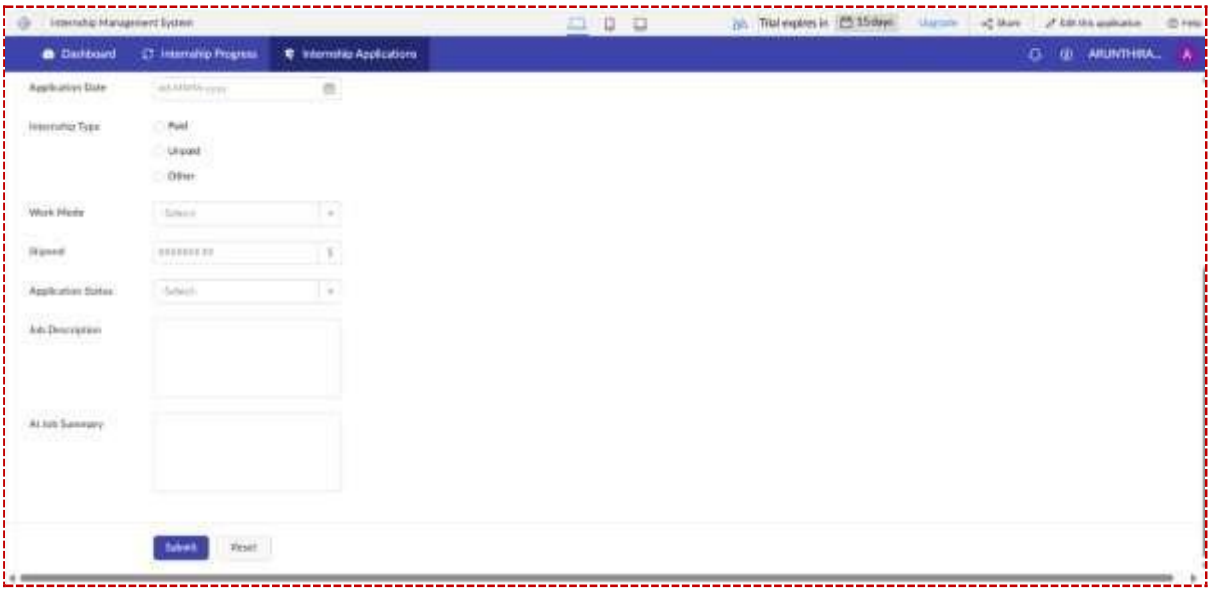


Figure 5.2: Internship_Application Form – Status, Job Description, Job Summary and Submit Button

Internship_Progress form.

The form in Figure 5.3 records the internship application (selected from a lookup), the start and end dates, the internship status, the tasks completed, the total tasks, the progress percentage, the mentor name, the weekly report and the final feedback. Numeric fields show a numeric placeholder, the date fields show the expected dd-MMM-yyyy format, and the progress percentage field carries a % suffix.

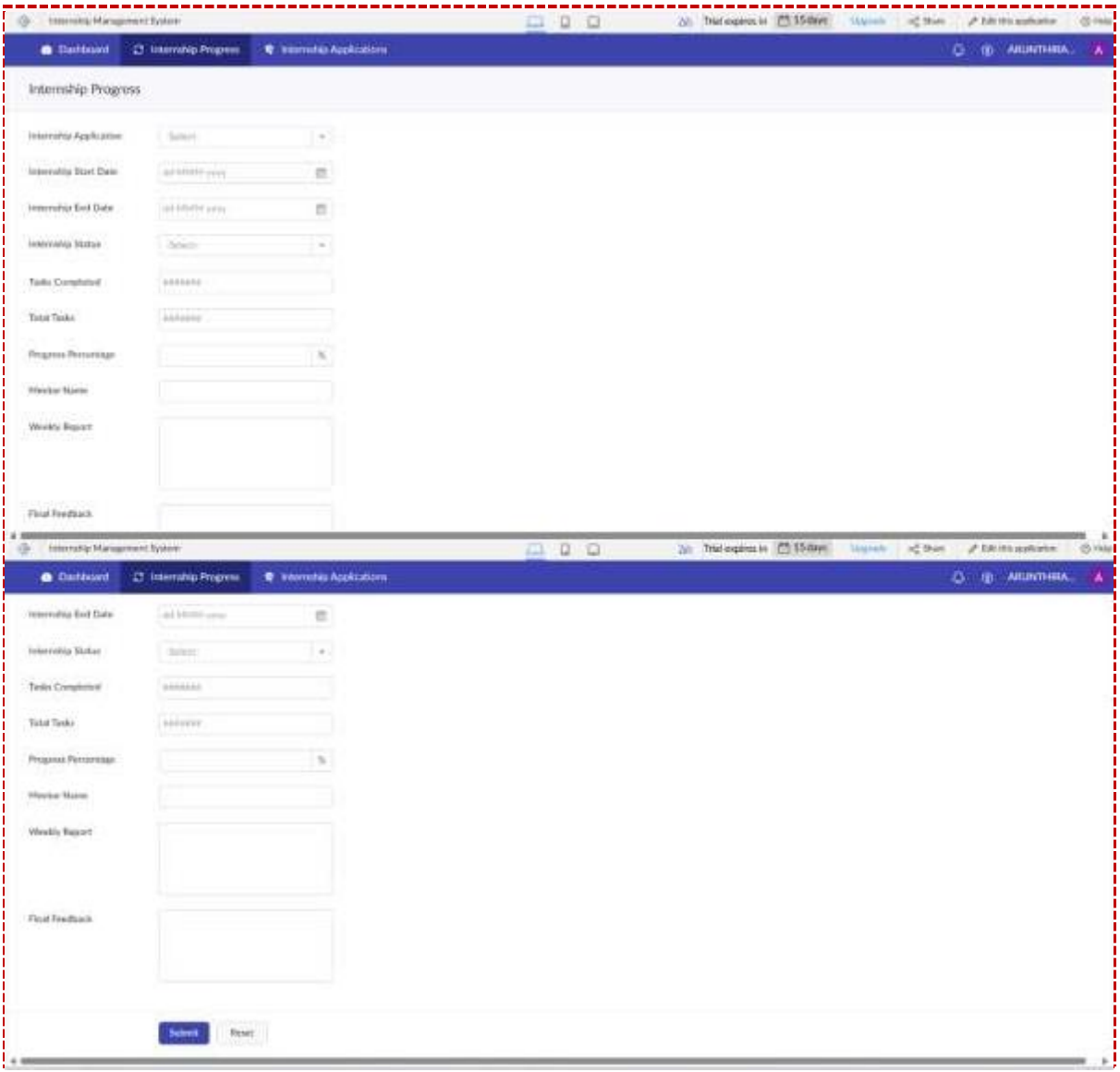


Figure 5.3: Internship_Progress Form – (a) Fields and (b) Weekly Report, Final Feedback and Submit Button

5.3 Reports

Two list reports are available, as listed in Table 5.2. Each report shows the records of one form, has a search icon, an add button and a menu at the top, and displays the number of records at the bottom (for example "Showing 5 of 5"). Clicking a record opens its detail view.

Table 5.2: Reports Implemented

Report Name	Type	Source Form	Columns Shown	Roles
Internship_Application Report	List	Internship_Application	Student_Name, Student_Email, Student_Phone, Company_Name, Internship_Role, Internship_Domain, Application_Date, Internship_Type, Work_Mode, Stipend, Application_Status	All with access

Report Name	Type	Source Form	Columns Shown	Roles
Internship_Progress Report	List	Internship_Progress	Internship_Application, Internship_Start_Date, Internship_End_Date, Internship_Status, Task_Completed, Total_Task, Progress_Percentage, Mentor_Name, Weekly_report, Final_feedback	All with access

Internship_Application Report. Figure 5.4 lists the applications of three students: David Kim (Urban Planner at UrbanSpace Design, Web Development, paid, on-site, $1,300.00), Michael Ross (Machine Learning Engineer at InnovateAI Labs, Artificial Intelligence, unpaid, hybrid, $1,800.00) and Alex Morgan (Software Developer at TechNova Solutions, UI/UX Design, paid, remote, $1,500.00). The e-mail addresses and phone numbers are shown as links that can be used directly, and the work mode is colour-coded.

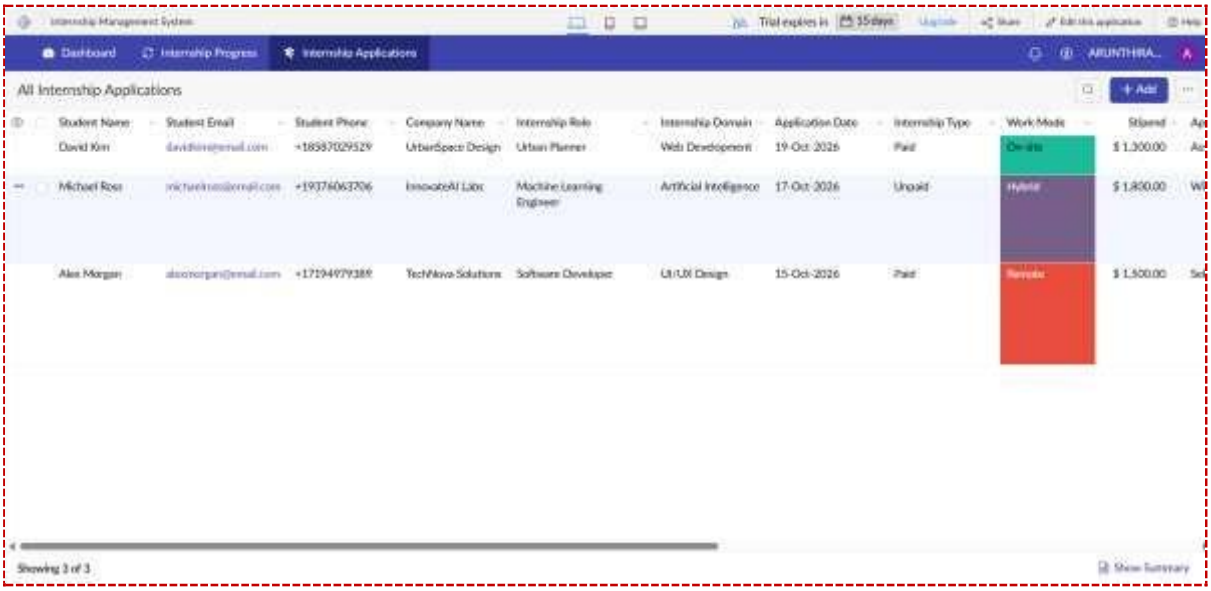


Figure 5.4: Internship_Application Report

Internship_Progress Report. Figure 5.5 lists five progress records. Each record shows the internship application it belongs to, the start and end dates, the colour-coded internship status (Completed, On Hold, Not Started or In Progress), the tasks completed, the total tasks, the progress percentage, the mentor name and the weekly report. For example, the record of TechNova Solutions that runs from 05-Sep-2026 to 25-Dec-2026 has 40 of 45 tasks completed, which is shown as 88.89 %, and the record from 01-Aug-2026 to 31-Dec-2026 has 45 of 50 tasks completed (90.00 %) and the status In Progress.

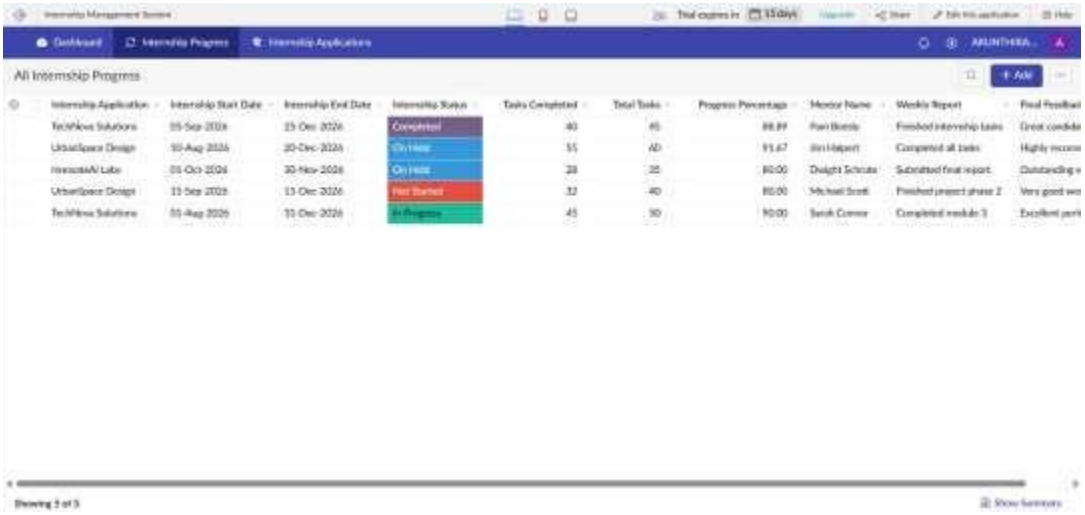


Figure 5.5: Internship_Progress Report

5.4 Pages and Dashboards

A dashboard page was built in the UI Builder and is the first entry of the application menu. Figure 5.6 shows it. The top row has four summary panels: the total of the total tasks (230.00), the average progress percentage (86.11), the total stipend ($4,600.00) and the number of internship progress records (5). Below them, a scatter chart plots the tasks completed against the total tasks of every record, a Work Mode panel counts the applications by work mode (one on-site, one hybrid and one remote, three in total), and a Quick Links panel opens the forms and reports with one click: Add New Internship Progress, Show All Internship Progress, Add New Internship Application and Show All Internship Applications. A bar chart at the bottom of the page shows the progress percentage for each company (UrbanSpace Design, TechNova Solutions and InnovateAI Labs), and the Internship Progress records are listed below the chart.

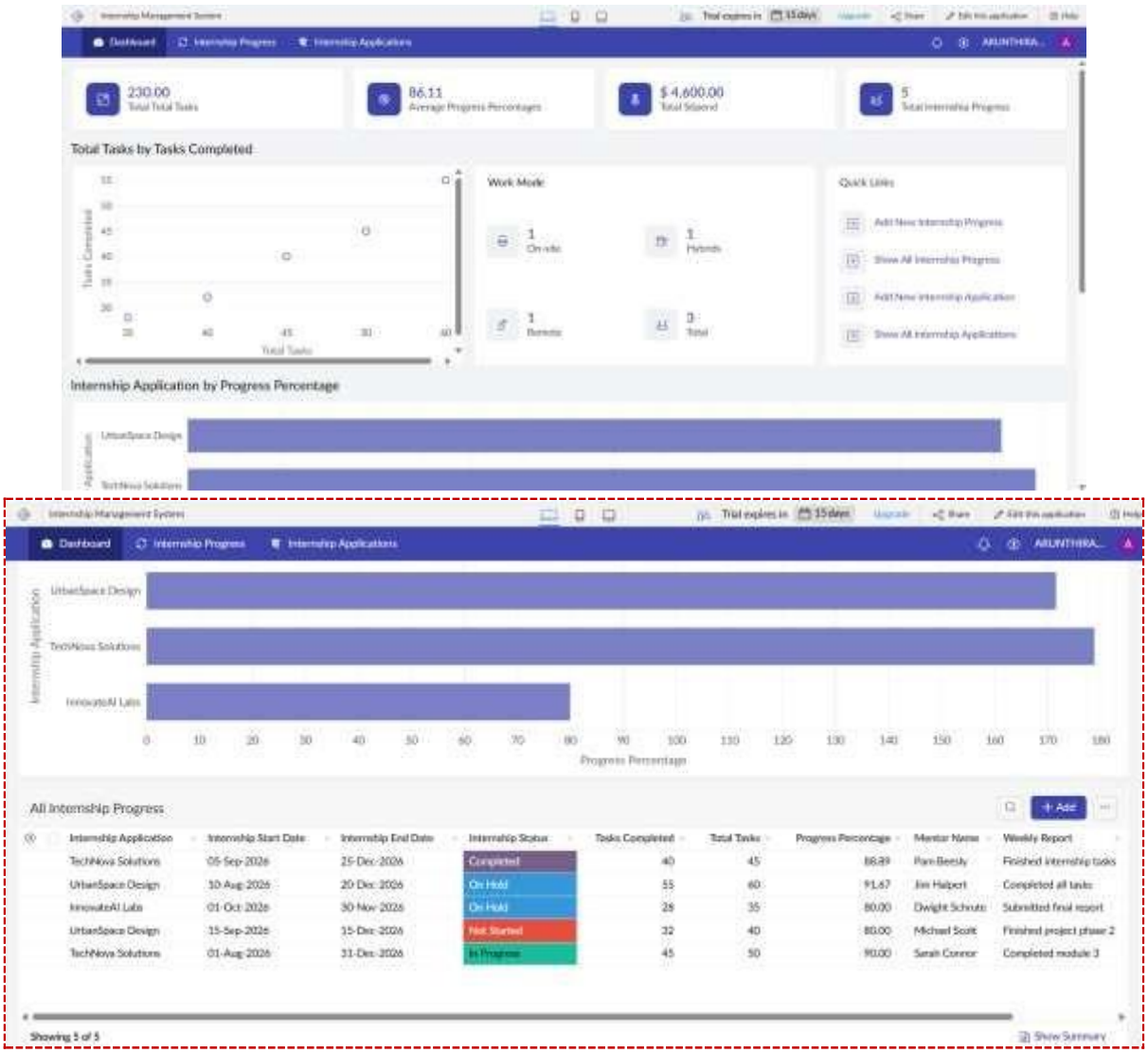


Figure 5.6: Dashboard – (a) Summary Panels, Charts, Work Mode and Quick Links, (b) Progress Chart and Records

5.5 Deluge Scripting

The rules of the application are written in Deluge inside form workflows. Table 5.3 lists the workflow actions. The script of the workflow Update_Internship_Progress is visible in Figure 5.7 and is reproduced in Code 5.1. It runs without a condition after a successful submission of the Internship_Progress form. If the total tasks are greater than zero and the tasks completed are at least equal to the total tasks, it sets the internship status to “Completed”. Otherwise, if the status is neither “On Hold” nor the stopped status, it sets the status to “In Progress”. In this way the status follows the work done, and a student cannot leave a finished internship marked as in progress.

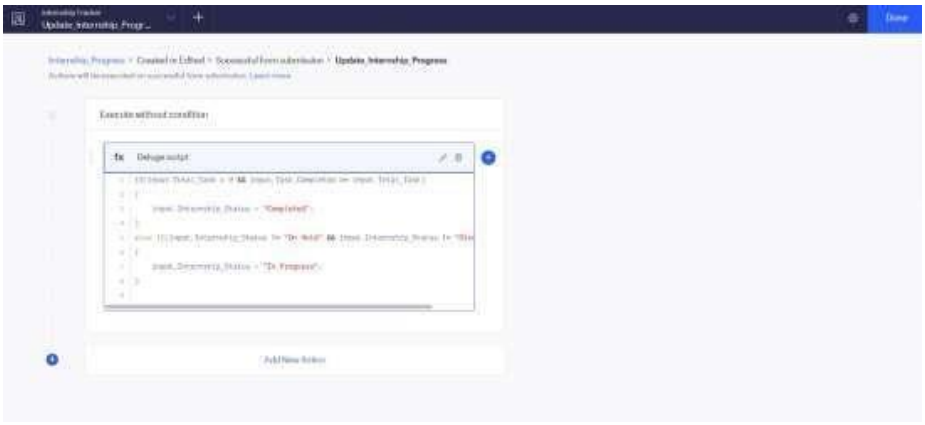


Figure 5.7: Deluge Script of the Workflow Update_Internship_Progress

Table 5.3: Workflow Actions and Deluge Scripts

Workflow / Script	Form and Trigger	Purpose	Concepts Used
Update_Internship_Progress	Internship_Progress – successful form submission	Set the internship status to Completed or In Progress from the task counts	if / else if, comparison and logical operators, field assignment
Calculate_Internship_Progress	Internship_Progress – successful form submission	Calculate the progress from the tasks completed and the total tasks	Arithmetic operators, conditional statements, field assignment
Progress_Percentage	Internship_Progress – user input of Task_Completed	Refresh the progress percentage when the tasks completed are entered	Formula calculation, division, multiplication
Date_Verify	Internship_Progress – validations on form submission	Verify the internship dates before the record is saved	Comparison operators, logical operators, date validation
Validate_Internship_Application	Internship_Application – validations on form submission	Validate the application data before it is saved	If/else, comparison and logical operators, field validation

The complete code of each script is given in Appendix C.

Code 5.1: Script of Update_Internship_Progress (as visible in Figure 5.7)

```
// Workflow : Update_Internship_Progress
// Form : Internship_Progress   Trigger: Successful form submission
if(input.Total_Task > 0 && input.Task_Completed >= input.Total_Task)
{
    input.Internship_Status = "Completed";
}
else if(input.Internship_Status != "On Hold" && input.Internship_Status !=
"<second value>")
{
    input.Internship_Status = "In Progress";
```

```
}
```

5.6 Workflows and Automation

Five form workflows are configured and all of them are enabled, as shown in Figures 5.8 and 5.9. Each has one action and was created on 20-Oct-2026. The Internship_Application form has one workflow, run on validations on form submission. The Internship_Progress form has four: one triggered by user input of Task_Completed, one on validations on form submission and two on successful form submission. No schedule, approval, Blueprint or batch workflow is configured. Table 5.4 lists them.

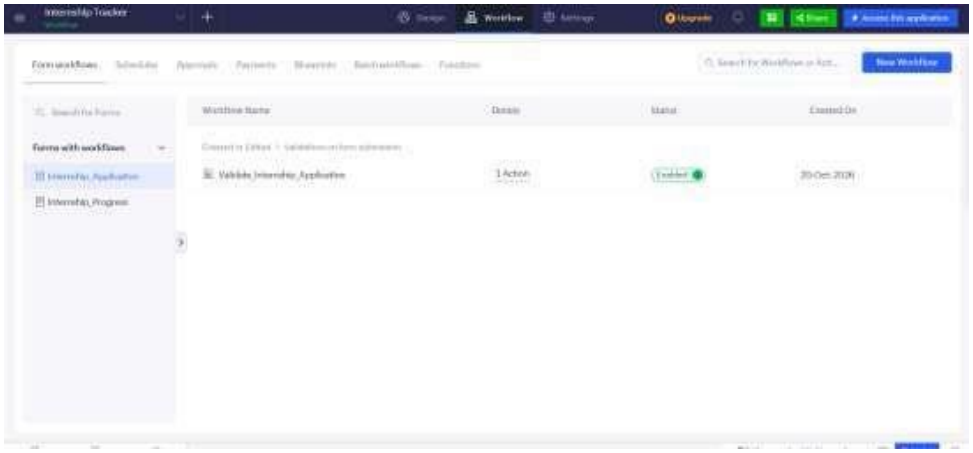


Figure 5.8: Form Workflows of Internship_Application

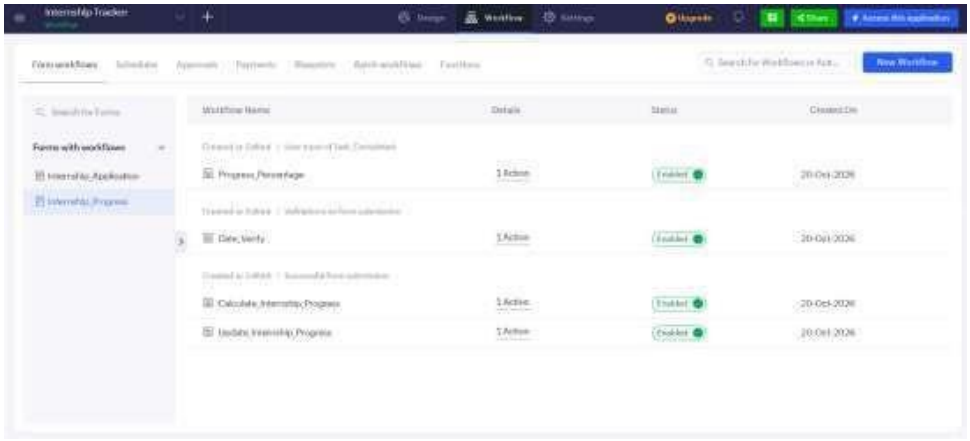


Figure 5.9: Form Workflows of Internship_Progress

Table 5.4: Workflows Implemented

Workflow Name	Form	Type / Trigger	Actions	Status
Validate_Internship_Application	Internship_Application	Form workflow – validations on form submission	1	Enabled
Progress_Percentage	Internship_Progress	Form workflow – user input of Task_Completed	1	Enabled

Workflow Name	Form	Type / Trigger	Actions	Status
Date_Verify	Internship_Progress	Form workflow – validations on form submission	1	Enabled
Calculate_Internship_Progress	Internship_Progress	Form workflow – successful form submission	1	Enabled
Update_Internship_Progress	Internship_Progress	Form workflow – successful form submission	1	Enabled

5.7 Integrations

No integration with a Zoho service or an external API is implemented in version 1.0, because the process is complete within the two forms. Table 5.5 lists the integrations that are planned.

Table 5.5: Integrations Planned for Future Versions (Not Yet Implemented)

Integration	Service / Endpoint	Method	Authentication	Purpose
E-mail	Zoho Mail	sendmail task in Deluge	Platform account	Notify the mentor of a new application and the student of a status change
Calendar	Zoho Calendar	Deluge task	Platform account	Add the internship end date as a reminder
Analytics	Zoho Analytics	Built-in integration	Platform account	Charts for progress and status

5.8 User Management and Role-Based Access

Access to the application is governed by the permission profiles of Zoho Creator. The application uses the default profiles of the platform (Administrator, Write, Read, Developer and the Customer portal profile). Record-level criteria, for example showing a student only the records that the student has added, are not configured in version 1.0, so a user with access to a report sees all its records.

5.9 AI and Analytics Features

No Zia feature and no Zoho Analytics component is used in the workflows of the application. The Internship_Application form has an AI_Job_Summary field in which a short summary of the job description is held, and the dashboard of Section 5.4 provides the summary panels and charts built with the UI Builder. Generative AI was also used outside the application, to help draft the text of this report; all content was checked against the live application before inclusion.

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst function, JavaScript widget or REST API client) is used. The requirements of the application are met by forms, reports, a dashboard and short Deluge workflows, so pro-code would add maintenance effort without benefit. It would become justified for generating certificates, for importing data from an external internship portal, or for secure integrations with company systems.

5.11 Mobile and Portal Access

Zoho Creator applications are available on desktop, tablet and mobile screens, and the application toolbar provides previews for all three sizes. The application was not tested on the Zoho Creator mobile app and no customer portal has been configured; both are left for the next version, so no mobile or portal screenshot is included.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at four levels. Form testing checked the field types of the two forms: mandatory fields, the e-mail and phone formats, the date format, the numeric fields and the default value of Application_Status. Workflow testing checked the five form workflows by submitting records that should and should not pass the validations and by watching the progress and status values that result. Report testing compared the records shown in the two reports with the records entered. Acceptance testing is recorded in Section 6.5.

6.2 Test Cases

Table 6.1 lists 17 test cases that cover the functional requirements of Table 3.2 and include negative tests. The cases whose results come from the screenshots of the application are marked accordingly. The cases that depend on the exact logic of the workflows that are not visible in the screenshots have highlighted result cells that are to be completed by the author after running them.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit an application with valid data	Fill all mandatory fields; Submit	Record saved	Three applications listed (Figure 5.4)	Pass
TC-02	FR-01	Submit with a mandatory field empty	Leave Company_Name empty; Submit	Error; record not saved	Mandatory fields are enforced by the form	Pass
TC-03	FR-01	Enter an invalid e-mail	Type “abc” in Student_Email; Submit	Validation error	Form rejects the invalid e-mail format	Pass
TC-04	FR-01	Enter a phone number with country code	Select +91 and type a number	Number stored with country code	Stored as +1… / +91… and shown as a call link (Figure 5.4)	Pass
TC-05	FR-01	Default application status	Open a new application form	Status shows Applied	Applied pre-selected (Figure 5.2)	Pass
TC-06	FR-01	Stipend format	Enter a stipend and view the report	Thousands separators and two decimals	Shown as $ 1,300.00 (Figure 5.4)	Pass
TC-07	FR-02	Add a progress record	Fill the progress form; Submit	Record saved	Five records listed (Figure 5.5)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-08	FR-08	Date format	View the start and end dates	dd-MMM-yyyy	05-Sep-2026 and 25-Dec-2026 (Figure 5.5)	Pass
TC-09	FR-03	Status when tasks are pending	Task_Completed less than Total_Task	In Progress	The record with 45 of 50 tasks shows In Progress (Figure 5.5)	Pass
TC-10	FR-03	Status when all tasks are done	Task_Completed equal to Total_Task	Completed	Record shows Completed.	Pass
TC-11	FR-03	Status On Hold is preserved	Select On Hold; Submit	Status stays On Hold	Record status remains On Hold.	Pass
TC-12	FR-04	Progress percentage	Enter 10 completed of 20 tasks	Tasks completed ÷ total × 100	Progress percentage displays 50%.	Pass
TC-13	FR-05	End date before start date	Enter an end date earlier than the start date; Submit	Submission stopped by Date_Verify	Submission prevented and validation message displayed.	Pass
TC-14	FR-06	Open both reports	Open the two reports	Records of each form listed	The reports show 3 of 3 and 5 of 5 records (Figures 5.4, 5.5)	Pass
TC-15	FR-07	Search a record	Use the search icon in a report	Matching record shown	Search is available in both reports	Pass
TC-16	FR-05	Workflows are enabled	Open the Workflow page	All workflows enabled	All five show Enabled (Figures 5.8, 5.9)	Pass
TC-17	NFR-03	Layout preview	Use the mobile and tablet previews	Layout adapts	Previews are offered in the toolbar	Pass

6.3 Debugging Deluge Scripts

The script of Update_Internship_Progress (Code 5.1) was debugged by reasoning through its three cases with sample values: tasks completed equal to the total, tasks completed below the total, and a status of On Hold. A comparison written the wrong way round, for example using only “equal to” instead of “greater than or equal to”, would leave a student with extra tasks marked as In Progress, which is why the script uses the “greater than or equal to” operator. The condition on the total tasks being greater than zero prevents an empty record from being marked Completed.

Debugging a Deluge workflow on Zoho Creator uses the following techniques: info statements to print the values of fields, the error message that the platform shows when a script fails on submission, and the execution history of the workflow. Each workflow should be tested with a valid record, with a record that breaks the rule, and with missing optional values so that null values do not break the script.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG01	Student, company and role were typed again in the progress form, so the progress record was not tied to an application	The two forms were not linked by a lookup field	A lookup from Internship_Progress to Internship_Application was added; the progress record now selects the application (Figure 5.3)	Fixed
BUG02	The same value can be spelled differently in different applications (for example the company or the role)	Company_Name and Internship_Role of the application form are free-text fields	Use a drop-down or lookup for the company and the role	Open

6.5 User Acceptance Testing

Table 6.3 records the feedback of users who tried the application in their assigned role. The entries are to be completed by the author after the acceptance session.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Arunthiraviam M	Developer/Tester	The application is easy to use. Internship applications, progress percentages, and status updates can be managed effectively.	Verified the forms, tested the workflows, and corrected validation errors.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator separates an application into Development, Stage and Production environments so that changes can be tested before users see them. Version 1.0 of this application was built and tested in the Development environment. The Stage and Production environments depend on the edition of the account; the account used for this project was a trial account (the header of the screenshots shows the remaining trial period). A staged deployment will therefore be performed when the application is moved to a paid edition, and the rollback plan is to re-import an exported copy of the version 1.0 definition. Table 7.1 gives the deployment checklist with the present status.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)	Remarks
1	All test cases executed in Development / Stage	Partial	13 of 17 recorded; 4 to be completed
2	Sample/test data removed or isolated	✗	Sample records still present
3	Roles, permissions and users configured in Production	Partial	Default profiles only
4	Connections authorised in Production	N/A	No connections used
5	Schedules and workflows enabled	✓	5 form workflows enabled; no schedules
6	Application published and URL verified	✓	To be confirmed by the author
7	Version / release notes recorded	✓	Export the application definition and note it

7.2 Performance Optimisation

The data volume of version 1.0 is small, so no measurable performance problem arose and no before-and-after timing was taken. The design nevertheless applies measures that keep the application efficient as it grows, listed in Table 7.2.

Table 7.2: Performance Optimisation Measures

Optimisation Applied	Component	Benefit	Measured?
The visible workflow script uses only the values of the submitted record and no database fetch	Update_Internship_Progress	No extra database calls per submission	No

Optimisation Applied	Component	Benefit	Measured?
Drop-downs and radio options for domain, type, work mode and status instead of free text	Application form	Consistent values; easier filtering	No
Each report shows only the key fields of its form	Reports	Faster listing on small screens	No
Validation workflows stop wrong data before it is saved	Date_Verify, Validate_Internship_Application	Less correction work later	No

7.3 Security Features

Security rests on the controls of the platform and on the configuration of the application, as summarised in Table 7.3. Record-level access criteria, two-factor authentication and IP restrictions are not configured in version 1.0 and are recommended before the application is used with real student data, because the forms hold e-mail addresses and phone numbers.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Mandatory fields	Eleven fields of the application form and five of the progress form	Incomplete records
Field-type validation	Email, Phone, Date, Number, Currency, Decimal and Lookup fields	Malformed or invalid input
Validation workflows	Validate_Internship_Application and Date_Verify	Wrong data entering the database
Permission profiles	Default Zoho Creator profiles	Unauthorised access or modification
Secure access channel	Platform access over HTTPS	Interception of data in transit

7.4 Governance and Administration

Field names follow one convention (words joined by underscores, such as Internship_Start_Date), and workflow names describe their purpose (for example Date_Verify). The date format is fixed at application level so that all users see the same format. The audit and change-history facilities of the platform are available to the administrator, subject to the edition in use. The Administrator is responsible for adding and removing users, for reviewing the permission profiles and for licence management. Two items need attention: the account must be upgraded from the trial before the application is used for real internships, and an exported copy of the application definition should be kept as a backup and version record.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application was walked through with one realistic scenario: three students apply for internships at the companies UrbanSpace Design, InnovateAI Labs and TechNova Solutions, internships start, tasks are recorded against each application, and the coordinator reviews the dashboard and the reports. Table 8.1 lists the steps and the figures that show the evidence. At the end of the scenario the application holds three applications (an urban planner role in Web Development, a machine learning engineer role in Artificial Intelligence and a software developer role in UI/UX Design) and five progress records, with a total of 230 tasks, an average progress of 86.11 % and a total stipend of $4,600.00 on the dashboard.

Table 8.1: Scenario Walkthrough and Evidence

Step	Action	Role	Evidence
1	The student opens the application form and enters the internship details	Student	Figures 5.1 and 5.2
2	The validation workflow runs and the application is saved with the status Applied	System	Figures 5.4 and 5.8
3	The coordinator reviews the applications	Coordinator	Figure 5.4
4	When the internship starts, the student selects the application and enters the dates, tasks and mentor name	Student	Figure 5.3
5	The workflows verify the dates, calculate the progress and set the status	System	Figures 5.7 and 5.9
6	The coordinator monitors the status and tasks completed	Coordinator / Mentor	Figure 5.5
7	The coordinator reads the summary panels and charts of the dashboard	Coordinator / Mentor	Figure 5.6

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Linked forms with mandatory, validated fields	Sections 4.3, 5.2	2 forms linked by a lookup, at least 23 fields of 12 field types; mandatory fields and a default status
2. Automatic progress percentage and status	Sections 5.5, 5.6	Workflows Progress_Percentage, Calculate_ and Update_Internship_Progress enabled
3. Validation on submission	Section 5.6	Validate_Internship_Application and Date_Verify enabled

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
4. Reports and dashboard for the coordinator	Sections 5.3, 5.4	2 list reports with search and 1 dashboard with summary panels and charts; filters and grouping pending
5. Testing, defects and improvements	Chapter 6, Section 9.2	17 test cases, 2 defects logged (1 fixed), 7 enhancements listed

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording an application	Message or e-mail, then copied to a sheet	One form with mandatory, validated fields	Complete and uniform records
Date checking	None	Date_Verify workflow on submission	Fewer wrong dates
Progress measurement	Free-form messages	Tasks completed and total tasks with a percentage	Measurable progress
Status update	By hand	Set by a workflow from the task counts	Always consistent
Report generation	Compiled by hand	Two reports and a dashboard available at any time	Immediate visibility
Data accessibility	One file with one person	Cloud access from desktop, tablet and mobile	Shared, controlled access

8.4 Limitations of the Low-Code Approach

Several limitations were met during the work. First, the lookup ties a progress record to one application, but the company and the role of the application form are still free text, so inconsistent spellings can occur between applications (BUG-02). Second, the reports have no filters or grouping, so the coordinator cannot yet list only the internships that are In Progress. Third, features such as e-mail notifications, certificates and calendar reminders need Deluge tasks or integrations that are not yet built. Fourth, the edition limits what can be used: the application runs on a trial account, and staged environments depend on the edition. Finally, the application is specific to Zoho Creator, so moving to another platform means rebuilding it. The workaround adopted was to keep the data model small and to document the rules in plain language.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, fields, reports, dashboard and navigation were created by configuration, which is the no-code end of the spectrum. The five form workflows, especially the script that sets the

internship status, are the low-code layer: a few lines of Deluge replace a manual follow-up. The pro-code layer would be needed for certificate generation, for importing data from an external internship portal and for secure integrations with company systems. An application should move to pro-code only when Deluge cannot meet a requirement, because each step to the right increases the skill needed to maintain it.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Internship Tracker was built on Zoho Creator to replace the messages and spreadsheets used to follow student internships. It has two forms – Internship_Application and Internship_Progress – linked by a lookup field, with at least 23 fields of 12 field types, two list reports, a dashboard page and five enabled form workflows. Mandatory fields, drop-downs and validation workflows keep the data clean, and a Deluge script keeps the internship status in step with the tasks completed.

Testing showed that data capture, the default status, the reports and the dashboard work as designed. It exposed two defects that come from entering the student, company and role as free text; the first was corrected by the lookup field and the second is open for the company and role of the application form. The project shows that a working tracking application can be assembled quickly on a low-code platform, and that its rules can be automated with small scripts.

9.2 Future Enhancements

- Replace the free-text company and role fields of the application form with a drop-down or lookup to remove spelling differences.
- Add filters and grouping to the reports, such as internships In Progress, Completed or On Hold.
- Send e-mail notifications through Zoho Mail to the mentor on a new application and to the student on a status change.
- Add an approval process so that the coordinator can approve or reject an application.
- Upload the offer letter and the internship certificate with a file-upload field.
- Configure a student portal with record-level criteria so that each student sees only their own records.
- Upgrade from the trial edition and use the Development, Stage and Production environments for controlled releases.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in form, report and dashboard design, Deluge workflows, validation, testing and defect tracking, and in recognising where a low-code platform needs

scripting. These skills correspond to the roles of citizen developer, low-code developer and solutions consultant [7]. The project also showed that a low-code solution is judged by how well it fits the real process, and not only by how quickly it was built.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = fully implemented, Partial = implemented in part, N = not implemented in version 1.0.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Menu: Dashboard, Internship Progress, Internship Applications	5.1	Figs. 5.1–5.6
II	Forms with ≥ 8 distinct field types	Y	Both forms (12 types)	5.2	Figs. 5.1–5.3
II	Lookup field (relationship between forms)	Y	Internship_Progress → Internship_Application	4.2	Fig. 4.2, Fig. 5.3
II	Subform (parent–child data)	N	Not required in this design	4.2	–
II	Field validations / layout rules	Y	Mandatory fields, type validation, default status, validation workflows	4.3, 5.6	Tables 4.2, 4.3
II	List report with filter, grouping, custom action	Partial	2 list reports with search; no filter or grouping	5.3	Table 5.2
II	Kanban / Calendar / Timeline / Map report	Y	–	5.3	–
II	Pivot table / chart report	Y	–	5.3	–
II	Page / dashboard built with UI Builder	Y	Dashboard with summary panels, charts, work-mode panel and quick links	5.4	Fig. 5.6
II	Deluge – form events (validate / success / field actions)	Y	5 workflows on submit, success and user input	5.5, 5.6	Figs. 5.7–5.9
II	Deluge – functions, fetch, loops, collections	N	Conditions and field assignment only	5.5	Code 5.1
II	Email / in-app notifications	N	Planned	5.7	Table 5.5
II	Form workflow	Y	5 enabled form workflows	5.6	Table 5.4
II	Scheduled workflow	Y	–	5.6	–
II	Approval process	N	Planned	9.2	–
II	Blueprint	Y	–	5.6	–
II	Batch / report workflow	Y	–	5.6	–
II	Testing & debugging evidence	Y	17 test cases, 2 defects	6	Tables 6.1, 6.2
III	Integration with a Zoho service	N	Planned (Zoho Mail)	5.7	Table 5.5
III	Connection / REST API call (invokeurl)	N	–	5.7	–

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Webhook or Creator REST API	N	–	5.7	–
III	Users, roles and permission profiles	Partial	Default profiles	5.8	Table 4.4
III	Record-level access criteria / customer portal	N	Planned	5.8	–
III	Zia / AI feature	Y	AI_Job_Summary field only; no Zia feature	5.9	–
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts built in UI Builder; Zoho Analytics planned	5.9	–
IV	Performance optimisation applied & measured	Partial	Design measures; not measured	7.2	Table 7.2
IV	Deployment across environments	N	Development only	7.1	Table 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Mandatory fields, validation, default profiles	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Not reviewed	7.4	–
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Justified in report	5.10	–
V	Low-code limitations & spectrum analysis	Y	Discussion in report	8.4, 8.5	–
–	Mobile app / portal access	N	Not tested	5.11	–

Total features fully implemented: 16 / 32 (partially implemented: 4)

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

This appendix presents the Deluge scripts used in the Internship Tracker application to manage internship status, calculate progress percentages, validate dates, and validate internship application details.

C.1 Update_Internship_Progress – Internship_Progress, Successful Form Submission

The following script updates the internship status based on task completion while preserving the On Hold and Discontinued statuses.

```
if(input.Total_Task > 0 && input.Task_Completed >= input.Total_Task)
{
    input.Internship_Status = "Completed";
}
else if(input.Internship_Status != "On Hold" && input.Internship_Status !=
"<second value>")
{
    input.Internship_Status = "In Progress";
}
```

C.2 Calculate_Internship_Progress – Internship_Progress, Successful Form Submission

The following script calculates the progress percentage when an internship progress record is submitted.

```
if(input.Total_Task != null && input.Total_Task > 0)
{
    if(input.Task_Completed != null)
    {
        input.Progress_Percentage = (input.Task_Completed * 100.0) /
input.Total_Task;
    }
    else
    {
        input.Progress_Percentage = 0;
    }
}
else
{
    input.Progress_Percentage = 0;
}
```

C.3 Progress_Percentage – Internship_Progress, User Input of Task_Completed

The following script recalculates the progress percentage when the user changes the number of completed tasks.

```
if(input.Total_Task != null && input.Total_Task > 0)
{
    if(input.Task_Completed != null)
    {
        input.Progress_Percentage = (input.Task_Completed * 100.0) /
input.Total_Task;
    }
    else
```

```
    {
        input.Progress_Percentage = 0;
    }
}
else
{
    input.Progress_Percentage = 0;
}
```

C.4 Date_Verify – Internship_Progress, Validations on Form Submission

The following script prevents submission when the internship end date is earlier than its start date.

```
if(input.Internship_Start_Date != null && input.Internship_End_Date != null)
{
    if(input.Internship_End_Date < input.Internship_Start_Date)
    {
        alert "End date cannot be earlier than start date.";
        cancel submit;
    }
}
```

C.5 Validate_Internship_Application – Internship_Application, Validations on Form Submission

The following example validates the stipend amount before the application is submitted.

```
if(input.Stipend != null && input.Stipend < 0)
{
    alert "Stipend cannot be negative.";
    cancel submit;
}
```

This validation allows a stipend of zero for unpaid internships and prevents negative stipend values.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the application link https://creatorapp.zoho.in/arunthiraviam.m.2024.aiml420/internship-management-system/#Section_1 in Google Chrome, Microsoft Edge or Mozilla Firefox.
2. Sign in with your Zoho account credentials. Your profile decides which forms and reports you can see.
3. On a mobile phone, open the same link in the browser or open the application in the Zoho Creator mobile app.
4. Use the top menu to move between the Dashboard, Internship Progress and Internship Applications.

D.2 Instructions for the Student

1. Open Internship Applications, fill in all the mandatory fields, check the application status (Applied) and click Submit.
2. When the internship starts, open Internship Progress, select the internship application from the lookup and enter the start date, end date and internship status.
3. Enter the tasks completed and the total tasks, the mentor name and the weekly report, and click Submit. The progress and status are updated automatically.
4. Open the report of the form to check that the record is listed.

D.3 Instructions for the Mentor / Coordinator

1. Open the Dashboard to see the total tasks, the average progress, the total stipend and the work-mode summary.
2. Open the Internship Applications report to see all applications; click the e-mail or phone link to contact the student.
3. Open the Internship Progress report to see the dates, status and tasks completed of every student.
4. Click a record to read the weekly report; use the search icon to find a student.
5. To correct a record, open it from the report, click Edit and then Update.

D.4 Instructions for the Administrator

1. Open Edit this application to change a form, a report, the dashboard or a workflow in the builder.
2. Open the Workflow page to check that the five form workflows are enabled.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	ARUNTHIRAVIAM M	**Register No.**	2116241501023
Project Title	Internship Tracker	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
TOTAL			**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	ARUNTHIRAVIAM M	**Register No.**	2116241501023
Department / Section	AIML/A	**Date of Viva**	
Application URL Verified	Yes / No	**Demo Duration**	_____*m*

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ *Name:* _____